inch), and the speed of the electrons' degeneracy motion there, 57 percent of the speed of light.

This electron speed was disturbingly large. Chandrasekhar, like R. H. Fowler before him, had computed the resistance of white-dwarf matter using the laws of quantum mechanics, but ignoring the effects of relativity. However, when any object moves at almost the speed of light, even a particle obeying the laws of quantum mechanics, the effects of special relativity must become important. At 57 percent of the speed of light, relativity's effects might not be too terribly big, but a more massive white dwarf with its stronger gravity would require a larger central pressure to support itself, and the random speeds of its electrons would be correspondingly higher. In such a white dwarf the effects of relativity surely could not be ignored. So Chandrasekhar returned to the starting point of his analysis, the calculation of the resistance to compression for white-dwarf matter, vowing to include the effects of relativity this time around.

To include relativity in the computation would require meshing the laws of special relativity with the laws of quantum mechanics—a mesh that the great minds of theoretical physics were only then working out. Alone on the steamer and barely graduated from university, Chandrasekhar could not produce that full mesh. However, he was able to produce enough to indicate the principal effects of high electron speeds.

Quantum mechanics insists that when already dense matter is compressed a bit, making each electron's cell smaller than it was, the electron's wavelength must decrease and correspondingly the energy of its degeneracy motion must increase. However, Chandrasekhar realized, the nature of the additional electron energy is different, depending on whether the electron is moving slowly compared to light or at close to light speed. If the electron's motion is slow, then, as in everyday life, an increase of energy means more rapid motion, that is, higher speed. If the electron is already moving at close to light speed, however, there is no way its speed can go up much (if it did, it would exceed the speed limit!), so the energy increase takes a different form, one unfamiliar in everyday life: The additional energy goes into inertia; that is, it increases the electron's resistance to being speeded up-it makes the electron behave as though it had become a bit heavier. These two different fates of the added energy (added speed versus added inertia) produce different increases in the electron's pressure, and thus different resistances to compression, Chandrasekhar deduced: at